Exact BdG evaluation: checks and returned information

1 Minimize	Compare $F_{\mathrm{impl}}(\Delta, q)$ minima and normal reference
2 Test numerical resolution	Check momentum-window enclosure and near-zero pairing amplitude
3 Separate status	Reliable label: Normal / uniform SC / FFLO Keep delicate and rerun flags separately
4 Test topology	For applicable FFLO states, evaluate bulk gap; only resolved gapped states receive $\nu = 0$ or 1

The surrogate schedules calls; it does not assign physical labels.